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Swathi Mohan

Personal portfolio
github.com/SwathiMohan
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Research interests: Formal methods, distributed systems, software testing, and reliable software systems.

Education

Manipal Institute of Technology, *BTech in Computer Science & Engineering (Minor: Data Science)* 2020—2024
CGPA: 9.26
Coursework: Formal Automata, Distributed Systems, Databases, Compiler Design, Operating System, Software Engineering
Madhava Kripa Higher Secondary School , *Class XII Aggregate: 95.2%* 2018-2020
Little Rock Indian School , *Class X Aggregate: 95.4%* 2018

Research Experience

SOFTWARE ENGINEER · MICROSOFT CoreAI (FoSSE) — July 2026 – Present

Research on specification-driven testing of distributed systems
Collaborators: Dr. Sriram Rajamani, Dr. Akash Lal, Dr. Aseem Rastogi

- **Specification Driven Harness for Production Cloud Services**
  TLA+, TLC Model Checking, Trace Validation, Fault Injection, C#
  - Generated end-to-end **fault-tolerance, concurrency and coverage tests** for Azure services from trace-validated TLA+ models, using model checking to select fault points and interleavings.
  - Designed a **property-preserving state-graph reduction algorithm** that merges model states the property under test cannot distinguish while keeping fault situations distinct (up to **147**$\times$ smaller; e.g. **117K** $\rightarrow$ **21K** states across 12 fault models), and planned covering test schedules over the reduced graph.
  - Found **22 bugs confirmed by service owners** (**6 high-severity**) across 5 services: silent partial failures such as duplicate execution, resurrected data and lock-safety violations.
  - Injected **13+ fault types** (lost responses, timeouts, crashes, lease expiry, slow storage) across **1,500+ fault interleavings**, checking recovery against specification properties.
  - Raised scenario coverage **8**$\times$ over existing suites; specification-action coverage **56%** $\rightarrow$ **95%** and branch coverage **29%** $\rightarrow$ **94%**, adding race and fault tests to services that previously had only mocked unit tests.

RESEARCH FELLOW · MICROSOFT RESEARCH (FoSSE) — July 2024 – June 2026

Advisors: Dr. Akash Lal, Dr. Aseem Rastogi, Dr. Pritam Gharat

- **Neuro-Symbolic Bug Finding and Repair for Rust**
  Static Analysis, Multi-Agent LLM Systems, MCP, Rust, C#
  - Built a multi-agent Rust checker: an orchestrator–subagent design exposing **12+ MCP tool endpoints**; static analysis proposes guideline violations (incl. panic, `unwrap`, `unsafe`), and LLM agents classify and fix them, validated by build and test.
  - Grounded agent decisions in **compiler-derived facts** (public API surface, visibility, AST structure), so every classification and fix respects type and API constraints rather than relying on LLM judgment alone.
  - Raised recall from **10–50% (LLM alone) to 80–95%** with **under 5% missed violations** across **300+ production crates (600K+ LoC)**.
  - Fixes **merged into Microsoft production code without manual edits**; featured in the **Rust@M365 newsletter**.
  - Added a pre-merge PR agent and an agent that synthesises new rule detectors from examples, accepted only with **zero missed violations** on ground truth.
- **LLM-Guided C $\rightarrow$ Rust Translation**
  Ownership Inference, Call-Graph Analysis, LLM Agents, MCP
  - Inferred pointer ownership over call graphs across **5K+ functions and 200K+ LoC**, summarising field-level ownership for **300+ C structs**.
  - Searched 4–6 competing ownership hypotheses per file with an LLM judge and backtracking: **+35% compile success**.
  - Extended the analysis to **lifetimes**, inferring borrow lifetimes from pointer usage so the translator emits safe references instead of raw pointers.
  - 5-stage translation agent on `zopfli`, `llama.c` and `tinyvm`: **~80% less manual annotation**, **70%+ compile success** without human intervention.

Work Experience

ML ENGINEER INTERN · FISCHERJORDAN **October 2023 – June 2024**

- **MyBrands — Invoice Understanding & Recommendation System**
  Python, Django, FastAPI, LangChain, FAISS, PaddleOCR, PostgreSQL, Docker, AWS
  - Built invoice understanding system (PaddleOCR + LLM extraction) processing **thousands of documents**; achieved **92%+ field extraction accuracy** in production.
  - Trained SVM (RBF) classifier for invoice email detection; **99%+ precision and recall**.
  - Designed hybrid RAG pipeline combining vector search (FAISS) with SQL reasoning: **90%+ answer accuracy**; containerised and deployed to AWS.
  - Shipped semantic recommendation engine: **+30% user engagement**, **+15% CTR**.
- **BrightQuery — Financial Intelligence Backend**
  Python, Django, PostgreSQL, Auth0, AWS
  - Solely designed and deployed Django backend serving **2M+ company records**; integrated Auth0 authentication and an LLM query layer over structured financial databases.
  - Optimised query handling for high-volume reads over a **200M+ company** dataset.

SOFTWARE ENGINEER INTERN · SAP LABS INDIA **June 2023 – July 2023**

- **Shift Roster Application**
  Vue.js, Flask, SQLite
  - Built end-to-end shift scheduling app (Vue.js frontend, Flask/SQLite backend) with **real-time roster updates**; improved scheduling accuracy **30%** and saved **15+ hrs/week**.
  - Overhauled API validation layer; cut system errors **40%** and increased uptime **20%**.

Technical Projects

Distributed File Sharding System **May 2023**

Python, Distributed System concepts

- Led the execution of a distributed file sharding system project in a team of 3, using client-server architecture and consistent hashing for fault-tolerant file partitioning and retrieval.
- **Achieved a sharding efficiency of 98%** by evenly distributing files across 10 servers using MD5 hash-based sharding.
- **Reduced data retrieval time by 75%** compared to a non-sharded system, with an average retrieval time of 3 milliseconds per file.
- Showcased adeptness in managing distributed systems, applying logical thinking, and utilizing Python proficiency.
- Link: Distributed File Sharding System

Document Quality Checker **Mar 2022 — Apr 2022**

HTML, CSS, Bootstrap, React js, Node js, MongoDB

- Developed a website employing AI/ML algorithms for document readability assessment and verification, including essential elements like photos, signatures, and dates.
- Achieved **Smart India Hackathon (SIH) 2022 finals qualification** by generating quality scores and enabling document uploads for authenticity verification.
- Link: Document Quality Checker

Skills

Formal Methods	TLA+, TLC model checking, trace validation, invariant inference, state-space reduction
Testing & Analysis	Fault injection, concurrency testing, coverage-guided test generation, static analysis (AST, call graphs)
Languages	Rust, C/C++, C#, Python, TypeScript
AI Systems	LLM agents, multi-agent orchestration, Model Context Protocol (MCP)
Infrastructure	Azure Storage / Cosmos DB emulators, Docker, PostgreSQL, AWS

Volunteering and Extra-Curriculars

- Placement Secretary, Manipal Institute of Technology, Manipal 2022
- Teaching Assistant, Manipal Institute of Technology, Manipal 2023–2024
  Assisted instruction for Data Structures & Algorithms and Compiler Design
- B.Tech Honours Program (Additional Academic Track), Manipal Institute of Technology 2021–2024
  Coursework: Advanced Data Structures & Algorithms, Quantum Computing, Machine Learning